

Exploiting Vision Language Model for Training-Free 3D Point Cloud Understanding via Improved Graph Score Propagation

Tiankai Chen, Yushu Li, Adam Goodge, Fei Teng, Xulei Yang, Zhen Liang, Tianrui Li, and Xun Xu

Abstract—3D point cloud understanding, e.g. out-of-distribution detection and recognition, remains a challenge, particularly in applications where safe and robust perception is critical. While recent 3D VLMs enable direct recognition via text prompts, their predictions are often unstable when the test distribution shifts or when only a handful of labeled samples are available. In this paper, we present **GSP++**, a graph-based inference framework that exploits the manifold structure of test-time point clouds to refine VLM scores without additional training. GSP++ constructs a graph over class prototypes, optional few-shot samples, and unlabeled test instances, and performs log-scaled score propagation to improve robustness across different dataset sizes while keeping computation efficient. The framework enables unified 3D OOD detection and classification. Moreover, we introduce a self-training strategy that selects high-confidence *positive* and *negative* samples and assigns them calibrated pseudo scores to further stabilize propagation. To enable practical deployment, we introduce dynamic graph construction and maintenance strategies that efficiently manage computation and memory, enabling scalable 3D understanding under both transductive and incremental inference. Extensive experiments on six datasets show that GSP++ not only reproduces and strengthens the full suite of OOD detection evaluations from GSP, a prior conference version, but also consistently improves zero-shot and few-shot 3D classification over strong VLM baselines. The code is available on <https://github.com/handsome999KK/GSP->

Index Terms—Computer Vision, Vision-Language Model, 3D Point Cloud, Graph, Out-of-distribution Detection, 3D Classification.

1 INTRODUCTION

Understanding 3D point cloud data is fundamental for the safe and reliable deployment of intelligent systems in real-world environments. For example, accurately recognizing and reasoning about surrounding objects is crucial for autonomous vehicles [2], [3] mobile robots to plan safely and avoid collisions [4], [5], [6]. Recent end-to-end autonomous-driving methods have further improved planning through reinforced diffusion [7], long-horizon world modeling [8], and momentum-aware reasoning [9]. Existing 3D understanding pipelines are often developed in a task-specific manner, relying on dedicated models and substantial labeled training data for each downstream scenario [10]. Such supervised approaches typically learn representations through reconstruction objectives [11], metric-based matching [12], or discriminative training strategies [13]. While effective, they require large-scale, carefully annotated datasets, which are expensive to collect for 3D sensing and may not cover the long tail of real-world variations. Moreover, models trained on limited distributions can struggle to generalize when encountering novel sensors, environments, or object appearances. These challenges motivate alternative 3D understanding solutions that reduce reliance on extensive labeled data collection while maintaining robust generalization in the wild.

Vision-language models (VLMs) have emerged as powerful foundation models that bridge visual features with textual understanding [14], [15], [16]. VLMs are trained to

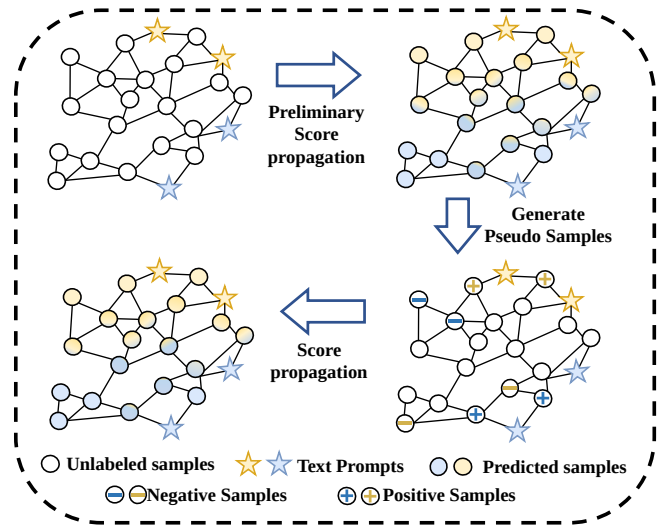


Fig. 1. Exploiting the testing data manifold for zero-shot 3D understanding. ID prototypes propagate scores to test samples, with confident ones selected as pseudo samples for refinement.

associate the content of visual inputs with their textual labels and they demonstrate strong zero-shot generalization to unseen visual content [14]. This capacity makes VLMs appealing for out-of-distribution (OOD) detection [17], [18], [19], [20] and classification [21], [22] without retraining. A common VLM-based OOD detection pipeline maps in-distribution (ID) class concepts, such as class names with various templates, into a shared feature space using the text

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† Correspondence to Xun Xu: xu_xun@i2r.a-star.edu.sg

encoder. The visual samples, similarly projected using the visual encoder, are then scored by comparing their distances to the ID class concepts in this feature space [17].

This VLM-based 3D understanding pipeline has shown competitive performance for 2D image detection, primarily due to the vast understanding of 2D VLMs from the massive-scale image-text pre-training datasets (e.g. 400M pairs for pre-training CLIP [14]). In contrast, VLMs for 3D point clouds, such as the state-of-the-art ULIP2 [23], are typically pre-trained on much more restricted datasets (e.g., 800k point cloud samples). This restricts their ability to generalize across unseen categories and distributional shifts (e.g., style changes), which reduces the effectiveness of similarity-based methods for 3D point cloud understanding.

In this work, rather than relying solely on the intrinsic capabilities of pre-trained 3D VLMs, we hypothesize that exploiting the geometric structure of unlabeled test samples can further improve their generalization ability. To this end, we introduce a graph-based approach that captures the underlying manifold of the test data and unifies 3D out-of-distribution (OOD) detection and classification [24], [25] within a single framework. As illustrated in Fig. 1, we first construct a graph over class prototypes (e.g., text prompts and optional few-shot 3D samples) and testing samples. We adopt a unified graph construction strategy with task-specific modifications. For OOD detection, prototype and few-shot nodes connect to their nearest testing samples, while testing samples are additionally connected through a KNN graph to capture the test-time manifold. For classification, each testing sample is linked to its nearest prototype and nearest few-shot labeled sample (if available), together with KNN connections among testing samples, yielding a lightweight yet class-aware propagation graph. Prototype nodes are then initialized with class-aware scores (e.g., one-hot vectors in multi-class classification or normalized OOD score for OOD detection), while testing samples start from neutral scores. We perform score propagation from prototype nodes to testing nodes for inference. The propagated scores are then used as class likelihoods for classification or ID/OOD confidence scores for OOD detection. Our framework seamlessly operates in both zero-shot and few-shot settings, consistently improving performance regardless of the availability of labeled samples.

We further introduce three designs to enhance the effectiveness of this framework. First, beyond straightforward score propagation on the graph [26], we notice an instability of hyper-parameters over varying datasets and graph sizes. Therefore, we introduce a log-scaled score propagation scheme to improve propagation stability across datasets and graph sizes. Second, while conventional training-free methods often average prompt features into a single prototype [24], [27], we argue that prompt templates exhibit non-trivial semantic diversity, and thus benefit from preserving multiple prompt modes. Finally, negative prompting has shown strong performance in VLM-based OOD detection [28], [29], [30], but directly negating text descriptions is often ineffective and learning negative prompts usually requires optimization. Instead, we employ self-training to convert high-confidence testing samples into pseudo-negative samples and assign them calibrated pseudo scores based on the initial propagated scores. Rather than using hard

pseudo-label assignment, we standardize the initial scores and apply a $\tanh$ mapping to obtain bounded soft values in $(-1, 1)$, so that selected samples provide confidence-aware supervision during the second-round propagation.

In many deployment scenarios, unlabeled test samples are available as a batch, enabling transductive inference, where the underlying test manifold can be jointly exploited for improved prediction. However, in practical online systems, test samples often arrive sequentially as a stream, making it impractical to access the entire manifold at once. To support efficient incremental inference, we incorporate a dynamic graph construction strategy that updates the graph on the fly using a lightweight neighborhood replacement mechanism, rather than rebuilding it whenever new samples arrive. Furthermore, to accommodate continuously growing test streams under bounded memory, we introduce a sliding-window graph maintenance strategy that keeps the graph size fixed while preserving effective semantic propagation. Together, these designs enable our framework to operate efficiently under both transductive and incremental inference settings.

We refer to the resulting framework as *Improved Graph Score Propagation (GSP++)*, reflecting its unified formulation for diverse 3D understanding tasks, enhanced graph score propagation and normalization, and support for scalable transductive and incremental inference. Our main contributions are summarized as follows.

- In the conference version [1], we employ a pre-trained vision-language model for 3D point cloud OOD detection and introduce Graph Score Propagation (GSP), a novel methodology that leverages the testing data manifold instead of solely relying on text-test feature distances.
- In the conference version [1], we enhance GSP with prompt clustering and self-training-based negative prompting. GSP also integrates few-shot ID training samples to further improve OOD detection accuracy.
- In this extension, we further propose **GSP++**, a more generic training-free graph-based adaptation framework for VLM-based 3D understanding, which enables both zero-/few-shot 3D classification and 3D OOD detection.
- We develop new components in GSP++, including log-scaled score propagation and a self-training pseudo-labeling strategy that selects high-confidence positive/negative test samples for more robust propagation. GSP++ also supports both transductive inference and dynamic graph expansion and maintenance for incremental inference.
- Extensive experiments on multiple 3D benchmarks demonstrate the effectiveness and flexibility of GSP++. In addition to reproducing and improving the original GSP results on 3D OOD detection, GSP++ achieves consistent improvements on zero-shot and few-shot 3D classification across ModelNet40 and ScanObjectNN, and remains effective under dynamic graph expansion, highlighting its practicality for scalable 3D understanding.

2 RELATED WORK

2.1 Vision-Language based OOD Detection

Traditional OOD detection methods mainly estimate model uncertainty from classification logits, with early approaches relying on softmax-based confidence scoring [13] and multi-scale global-local modeling [31]. With the emergence of vision-language models, OOD detection has also been studied through the lens of text prompting, where textual descriptions help separate in-distribution (ID) and OOD samples [18], [19], [20], [29], [30]. Wang et al. [19] propose CLIPN for zero-shot OOD detection by teaching CLIP to model negation semantics with a learnable “no” prompt and a dedicated text encoder, so OOD samples can be identified via their mismatch with ID prompts and alignment with “no” prompts. LoCoOp [20] leverages CLIP’s local patch-level features to perform OOD-aware regularization during prompt learning, encouraging the model to better separate in-distribution samples from out-of-distribution ones in the few-shot setting. Bai et al. [18] improve few-shot OOD detection by mining ID-like outliers near few-shot ID samples in CLIP feature space and learning ID-like prompts to better distinguish these hard, near-ID OOD examples, which benefits both OOD detection and few-shot learning [32]. Further improvements have been obtained by training text encoders [20] and adopting learnable prompts [29], [30] to enlarge the score margin between ID and OOD data. In parallel, recent efforts have extended OOD detection to 3D datasets and point clouds [33], [34]. In particular, 3D OOD detection has been recognized as a critical component for safety in autonomous driving applications [6], [35]. These developments motivate more effective VLM-based 3D OOD methods that can exploit both semantic priors from text prompts and structural information from test-time 3D data.

2.2 3D Vision-Language Models Classification

Vision-language models (VLMs) have recently become a strong foundation for open-vocabulary 3D classification by enabling direct matching between 3D representations and natural language descriptions. Early attempts bridge 3D and language by converting point clouds or meshes into multi-view images and then leveraging CLIP-style image-text alignment [21], [22], [36]. By inheriting semantic knowledge from large-scale image-text pre-training, these rendering-based methods can recognize unseen categories in a zero-shot manner and have demonstrated promising results on standard 3D benchmarks. CLIP2Point [37] transfers CLIP to point cloud classification through image-depth pre-training, using paired RGB and depth signals as intermediate supervision to better connect 2D visual semantics with 3D geometric structure. More recent approaches pursue direct 3D vision-language alignment to avoid the rendering step and learn unified representations across images, text, and point clouds. ULIP and its extensions [10], [23] align a point-cloud encoder with language embeddings through contrastive objectives, enabling zero-shot 3D classification via similarity between 3D features and text prompts. Beyond ULIP-style alignment, scaling-oriented pre-training methods such as OpenShape and Uni3D further improve open-vocabulary 3D recognition by exploiting large-scale cross-modal supervision and more diverse 3D-text/image

pairs, showing that data scale and modality diversity are critical for robust zero-shot transfer [38], [39]. In parallel, multimodal representation learning frameworks such as Point-Bind extend alignment to richer modality sets (e.g., image, text, audio, and point clouds), which can provide stronger transferable semantics for downstream 3D classification and retrieval [40]. This line of work highlights that properly designed pre-training data, scaling strategies, and prompt engineering can substantially strengthen zero-shot transfer [41]. In parallel, a number of studies explore stronger 3D encoders, improved cross-modal objectives, and richer textual supervision to further boost recognition performance and robustness.

2.3 Label Propagation

Label propagation (LP) is a classical paradigm in semi-supervised learning that infers labels for unlabeled samples by propagating supervision over a similarity graph constructed from labeled and unlabeled data [26], [42], [43], [44], [45], [46]. Its effectiveness is rooted in the smoothness (or cluster/manifold) assumption, i.e., nearby samples on the graph are likely to share the same label, which makes LP particularly suitable for transductive settings where test samples are jointly available at inference time [26], [42], [43]. Beyond classical graph-based SSL, later works showed that propagation can also be integrated with modern graph learning pipelines, e.g., by decoupling prediction and propagation or by applying LP-inspired post-hoc correction and smoothing to improve robustness and decision boundaries on graph-structured data [47], [48], [49]. In particular, GNNSafe leverages an energy-based OOD score from GNN classifiers and further enhances node-level OOD detection with a learning-free energy belief propagation scheme on graphs [49]. Recently, LP has also been introduced into vision-language model (VLM) adaptation and open-vocabulary recognition. ReCLIP [50] uses LP in an aligned vision-language projection space to generate pseudo labels from text prototypes to unlabeled target images, and then performs cross-modality self-training for source-free adaptation. ZLaP [24] tailors LP to bimodal graphs of text and image features for zero-shot classification, emphasizing geodesic-distance-based classification and efficient inductive inference via a dual formulation with sparsification. ECALP [25] further extends this line by dynamically constructing and expanding graphs over text prompts, few-shot samples, and test samples, together with context-aware feature re-weighting and iterative LP, yielding a unified training-free framework for zero-/few-shot adaptation and strong performance on fine-grained recognition and OOD generalization. Different from existing VLM+LP approaches that typically assume a closed set of category prototypes, our setting requires robust prediction and score refinement with limited class descriptions. To this end, we develop a self-training based negative prompting strategy and a prompt clustering scheme that enrich the prototype space and make graph-based propagation more reliable across diverse downstream tasks.

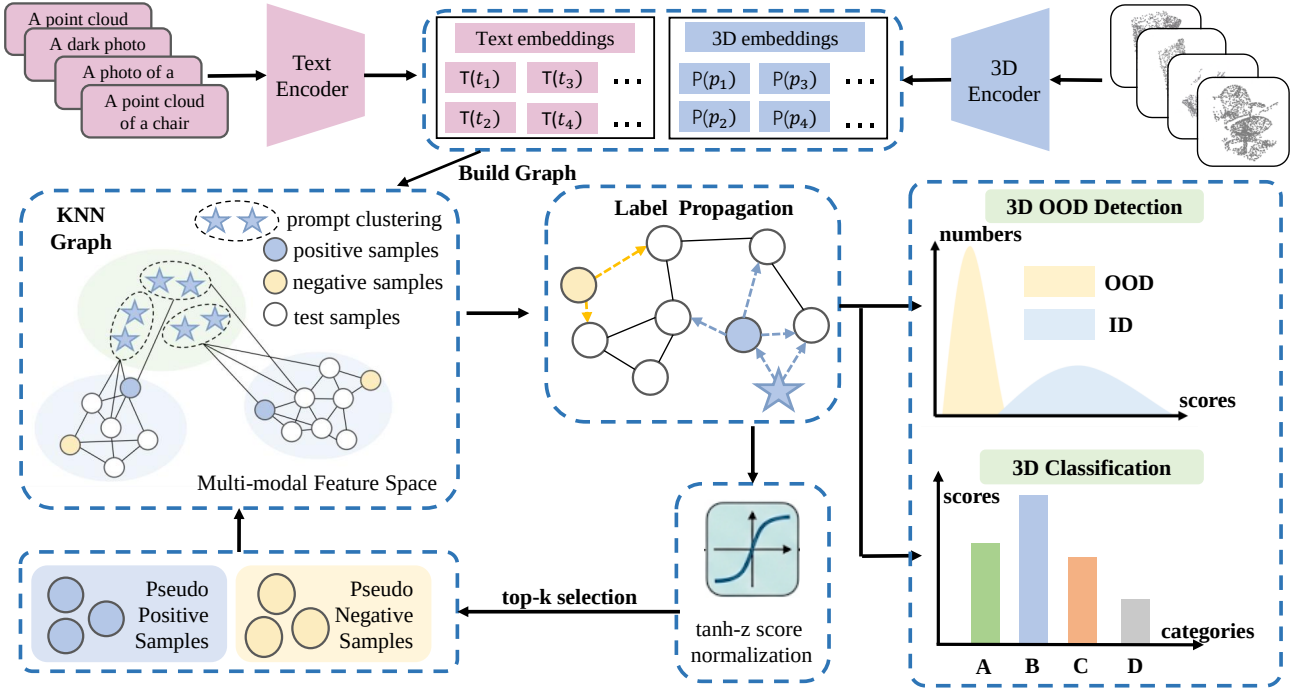


Fig. 2. Overview of the proposed framework for zero-/few-shot 3D understanding via graph score propagation. The framework encodes category prompts and dataset samples using frozen text and 3D encoders, constructing a KNN graph. Initial score propagation generates pseudo prompts, which refine the graph before final score propagation to compute final results.

3 PROPOSED APPROACH

3.1 Problem Formulation

We begin by formally defining the task of VLM-based 3D zero-/few-shot understanding. Let $z^f = f(x; \theta)$ and $z^t = g(t; \phi)$ denote the encoded features of point cloud data x and text t , where $f(\cdot; \theta)$ and $g(\cdot; \phi)$ are the VLM visual and text encoders respectively. The task involves the encoded features of unlabeled test data $\mathcal{D}_u = \{f(x_i)\}_{i=1 \dots N_u}$ and, optionally, the features and labels of few-shot labeled data $\mathcal{D}_l = \{f(x_j), y_j\}_{j=1 \dots N_l}$. The set of test samples' semantic classes is $\mathcal{Y} = \{1, \dots, C\}$, e.g. "Airplane", "Guitar", etc.. Following established practices for VLM adaptation [17], we represent each test class with N_t prompt templates $\mathcal{T} = \{t_j(y)\}_{N_t}$, e.g., "This is a point cloud of $\{y\}$." or "A good point cloud of $\{y\}$ ". These textual prompts are encoded by the text encoder, creating prototypes for all classes $\mathcal{P}$. Existing works often create prototypes by averaging the encoded prompts for each class, resulting in a prototype z_c^p for the c -th class, and the full set of prototypes is denoted $\mathcal{P} = \{z_c^p\}_{1 \dots C}$.

3.2 Vanilla 3D VLM Understanding

3.2.1 Preliminaries of 3D VLM

We briefly review the training objective of 3D vision-language models (VLMs). 2D VLMs are often the base model for 3D VLMs, which typically aligns visual features encoded via vision encoder and text features [51] via contrastive loss.

$$\mathcal{L}_{vlm} = \sum_i \log \frac{\exp(z_i^f z_i^t / \tau)}{\sum_j \exp(z_i^f z_j^t / \tau)} \quad (1)$$

On top of 2D VLMs, 3D VLMs introduce an additional 3D encoder to encode 3D point cloud and aligns 3D embedding towards 2D image and text embeddings via contrastive loss [10], [23], [37]. When a VLM is trained on a large amount of point cloud-text pairs, the cosine similarity of positive pairs (a point cloud sample and text prompt of its ground-truth class) is higher than that of negative pairs. Therefore, a pre-trained 3D VLM provides a simple yet effective baseline for 3D understanding: we encode 3D point cloud features with 3D encoder and use cosine similarity to quantify semantics.

3.2.2 Vanilla 3D VLM for OOD Detection

Pre-trained VLMs offer an effective approach to OOD detection by measuring the distance between visual samples and text labels in the feature space. For example, [17] employs the maximum normalized cosine similarity as the OOD score. We define a normalized score S to indicate OOD-ness, where $S \rightarrow 0$ suggests OOD and $S \rightarrow 1$ suggests ID.

$$S = \max_c \frac{\exp(-d(z^f, z_c^p) / \tau)}{\sum_j \exp(-d(z^f, z_j^p) / \tau)} \quad (2)$$

It is assumed that in the feature space, OOD samples are further from the ID prototypes than ID samples. Subsequent research has further extended the vanilla approach by introducing techniques like negative prompts to better separate OODs from ID samples [52].

3.2.3 Vanilla 3D VLM for Classification

VLMs also provide an effective baseline for zero-/few-shot 3D classification by comparing the feature of a point cloud with text prototypes. Given a test sample feature

$z_i^f = f(x_i; \theta)$ and the prototype set $\mathcal{P} = \{z_c^p\}_{c=C_1}^{C_{in}}$ constructed from prompt templates $\mathcal{T}$, the predicted class is obtained by maximum similarity matching:

$$\hat{y}_i = \arg \max_{c \in \mathcal{Y}_{in}} \text{sim}(z_i^f, z_c^p), \quad (3)$$

where $\text{sim}(\cdot, \cdot)$ is typically cosine similarity. For evaluation and confidence estimation, the similarities can be converted into class probabilities with a temperature-scaled softmax operation:

$$p(y = c | x_i) = \frac{\exp(\text{sim}(z_i^f, z_c^p)/\tau)}{\sum_{k=1}^{C_{in}} \exp(\text{sim}(z_i^f, z_k^p)/\tau)}. \quad (4)$$

In the few-shot setting, $\mathcal{D}_l$ provides additional supervision that can be used to refine the prediction rule or calibrate the similarities, while keeping the same text prototypes as semantic anchors.

The vanilla OOD detection or classification by 3D VLM achieves impressive results by making predictions on testing samples individually. As a result, it can be sensitive to real-world issues like domain/style shift, high/low intra/inter-class variations, background noise, etc. This motivates approaches that further exploit the manifold structure of the unlabeled test set $\mathcal{D}_u$ and optionally incorporate reliable few-shot clues to further improve robustness beyond direct similarity matching.

3.3 Modeling Manifold via Graph Construction

Instead of directly measuring the distance for OOD detection/ classification, we formulate 3D understanding by considering the manifold of unlabeled testing data $\mathcal{D}_u$. Specifically, the manifold can be captured by a sparse graph constructed from data samples where the affinity/adjacency characterizes the visual similarity. Each node on the graph represents either an unlabeled testing sample, a few-shot labeled sample, or a prototype. Below, we introduce a standard practice of building a graph catered for OOD detection and classification.

3.3.1 Graph for OOD Detection

We introduce the approach for constructing a graph for OOD detection. As observed by [24], [25], the distance between text prompts and visual samples is generally larger than the distance between visual samples. This observation also holds for 3D point cloud VLM because the 3D VLMs are often finetuned from 2D VLMs [10], [23]. Therefore, we adopt a similar approach to construct the graph. We write the weight matrix W in a blockwise form as shown in Eq. 5, where $W_u \in \mathbb{R}^{|\mathcal{D}_u| \times |\mathcal{D}_u|}$, $W_{up} \in \mathbb{R}^{|\mathcal{D}_u| \times |\mathcal{P}|}$ and $W_{ul} \in \mathbb{R}^{|\mathcal{D}_u| \times |\mathcal{D}_l|}$. No connections are created between prototypes and few-shot labeled samples. All blocks are further constructed as a k-nearest neighbor (KNN) graph following [24], [25]. It is also computationally efficient to adapt an existing graph to new testing samples by incrementally adding the testing samples to the graph [25].

$$W = \begin{bmatrix} \mathbf{I}^{|\mathcal{P}| \times |\mathcal{P}|} & \mathbf{0} & W_{up}^\top \\ \mathbf{0} & \mathbf{I}^{|\mathcal{D}_l| \times |\mathcal{D}_l|} & W_{ul}^\top \\ W_{up} & W_{ul} & W_u \end{bmatrix} \quad (5)$$

3.3.2 Graph for Classification

For 3D classification task, different from the OOD setting where each prompt/labeled sample connects to its k nearest test samples, classification requires propagating class-specific information to each individual test instance. Therefore, we construct a sparse bipartite linkage from each test sample to its most relevant supervision nodes. Concretely, for every unlabeled test feature in $\mathcal{D}_u$, we retrieve (i) the nearest text prototype in $\mathcal{P}$ and (ii) the nearest few-shot labeled sample in $\mathcal{D}_l$ (if available), measured by cosine distance in the VLM feature space. This results in at most one edge from a test node to the prototype set and at most one edge from a test node to the labeled set, ensuring a lightweight and class-aware assignment of supervision signals. Meanwhile, the unlabeled test nodes are connected among themselves by a standard k -nearest-neighbor (KNN) graph to capture the local manifold structure.

Following the same blockwise notation as Eq. 5, we write the weight matrix W for classification as

$$W = \begin{bmatrix} \mathbf{I}^{|\mathcal{P}| \times |\mathcal{P}|} & \mathbf{0} & W_{up}^\top \\ \mathbf{0} & \mathbf{I}^{|\mathcal{D}_l| \times |\mathcal{D}_l|} & W_{ul}^\top \\ W_{up} & W_{ul} & W_u \end{bmatrix}, \quad (6)$$

3.4 Unified 3D VLM Understanding as Score Propagation

Instead of directly measuring the distance for OOD detection/ classification, we formulate 3D understanding as a score propagation problem. Specifically, we assign a positive score (+1) to text prompts $\mathcal{P}$, while all testing samples are initially assigned a neutral score (0). Labels are then propagated from the positive nodes to all testing samples in an iterative manner. We follow a standard propagation rule [53] as shown in Eq. 7,

$$S_t = \tilde{W} S_{t-1} + \alpha S_0, \\ \text{s.t. } \tilde{W} = D^{-\frac{1}{2}} W D^{-\frac{1}{2}}, \quad D = \text{diag}(\sum_j W_{ij}), \quad (7)$$

$$S_0 = [S_0^p, S_0^l, S_0^u],$$

For different task, S_0 represents different initial scores and follows the constraints below:

$$\text{OOD Detection: } S_0^p = \mathbf{1}^{p \times 1}, S_0^l = \mathbf{1}^{l \times 1}, S_0^u = \mathbf{0}^{u \times 1}$$

$$\text{Classification: } S_0^p = \text{diag}(\mathbf{1}^{|\mathcal{P}|}), S_0^l \in \{0, 1\}^{|\mathcal{D}_l| \times C}, \quad (8)$$

$$S_{0ic}^l = \mathbb{1}(s_i = c), S_0^u = \mathbf{0}^{|\mathcal{D}_u| \times C}$$

Each step of propagation updates the score for unlabeled testing samples as a weighted average of scores from adjacent nodes and initial scores S_0 until T iterations are completed. Consequently, nodes that are closer to the prototypes on the manifold will obtain a higher score.

3.4.1 Log-Scaled Iterations for Label Propagation

Efficient solution to Eq. 7 requires multiple steps of iteration. Since different test datasets contain varying numbers of samples, a fixed propagation depth may either be insufficient for large graphs or wasteful for small ones. To ensure adequate propagation while avoiding unnecessary computation, we set the number of propagation iterations to

$$T = \lceil \log N_u \rceil, \quad (9)$$

where N_u is the number of test samples (unlabeled nodes) in the transductive graph. This choice is motivated by classical properties of diffusion and random-walk processes on similarity graphs. Each LP update can be viewed as a normalized message passing step that expands the effective receptive field by approximately one hop. For kNN-style similarity graphs, it is common to observe small-world behavior, where typical shortest-path lengths scale logarithmically with the number of nodes, implying that $\mathcal{O}(\log N_u)$ steps are sufficient to reach a global communication scale without overly aggressive smoothing [54]. In addition, LP is closely related to (lazy) random walks and diffusion with restart on the normalized adjacency; the relaxation/mixing time of such Markov processes on well-connected graphs is known to scale as $\mathcal{O}(\log N)$ under standard spectral conditions (e.g., expander-like connectivity) [55], [56]. Finally, classic LP formulations admit a unique fixed point and are often computed by iterative schemes whose error decays geometrically with the iteration count, suggesting that a logarithmic iteration budget provides a principled accuracy–efficiency trade-off for large test graphs [42], [57], [58]. A detailed graph-scale communication analysis supporting this logarithmic propagation depth is provided in the supplementary material.

3.4.2 Prompt Clustering

The above propagation strategy propagates labels from prototypes to testing samples on the graph. Thus, it is critical to design effective prototypes that capture the diversity of textual descriptions for all classes. Contrary to existing practices that average the text prompt features across all prompt templates, we propose modeling the distribution of prompt templates with multiple representative ones. Given the prompt templates $\mathcal{T}$, we first initialize a pool of text prompts $\mathcal{P}_c = \{z_{cj}^p = g(t_j(y_c))\}_{j=1 \dots |\mathcal{T}|}$ for the c -th class. We then introduce a clustering step to reduce this set to N_c representative prototypes $\tilde{\mathcal{P}}_c$. This is achieved by applying K-means [59] clustering on $\mathcal{P}_c$, and the cluster centers are adopted as the ID prototypes:

$$\begin{aligned} \tilde{\mathcal{P}}_c &= \arg \min_{\{\mu_k\}, \gamma} \sum_{N_c} \sum_{z_{cj}^p \in \mathcal{P}_c} \gamma_{jk} \|z_{cj}^p - \mu_k\|^2, \\ \text{s.t. } \gamma &\in \{0, 1\}^{|\mathcal{T}| \times N_c}, \sum_k \gamma_{jk} = 1, \end{aligned} \quad (10)$$

where μ_k are the cluster centers, and γ_{jk} indicates the assignment of each prompt feature z_{cj}^p to its closest center. The new prototypes, $\tilde{\mathcal{P}} = \tilde{\mathcal{P}}_1 \cup \tilde{\mathcal{P}}_2 \cup \dots \tilde{\mathcal{P}}_C$, are used for score propagation as positive nodes.

3.4.3 Pseudo-Labeling via Self-Training

Negative prompts have been shown to substantially benefit OOD detection [28], [29], [30]. Incorporating self-training signals is particularly straightforward within the graph label propagation framework, since high-confidence test samples can be re-used to refine the initialization of propagation. Existing approach exploits ID data to learn negative prompt [28] which prohibits zero-shot inference. To address this, we propose a self-training strategy that selects *positive samples* and *negative samples* from the test set and assigns them *soft pseudo-label scores* for the next propagation stage,

as illustrated in Fig. 1. Specifically, we first conduct an initial score propagation for T iterations following Eq. 7, giving rise to initial propagated scores S_T for all test samples. For the OOD detection task, we select the top- m and bottom- m samples from S_T :

$$\begin{aligned} \mathcal{I}_{pos} &= \text{Top-}m(S_T), \\ \mathcal{I}_{neg} &= \text{Bottom-}m(S_T), \end{aligned} \quad (11)$$

where $\text{Top-}m(\cdot)$ and $\text{Bottom-}m(\cdot)$ return the indices of the m largest and m smallest values, respectively. For the classification task, we perform class-wise selection: for each class $c \in \{1, \dots, C\}$, we select top- m and bottom- m samples according to the class score in S_T , and denote the selected sets by $\mathcal{I}_{pos}^c$ and $\mathcal{I}_{neg}^c$, respectively. Instead of assigning hard scores $+1$ and -1 [1], we compute a calibrated pseudo-label score for each selected sample based on S_T . We first compute the global mean $\mu = \text{mean}(S_T)$ and standard deviation $\sigma = \text{std}(S_T)$, and apply Z-score normalization followed by a tanh mapping:

$$z = \frac{S_T - \mu}{\sigma}, \quad S' = \tanh(z). \quad (12)$$

We then re-initialize the propagation scores using S' on the selected positive/negative samples:

$$\tilde{S}_0[i] = \begin{cases} S_0[i] & \text{if } i \notin \mathcal{I}_{pos} \text{ \& } i \notin \mathcal{I}_{neg}, \\ S'[i] & \text{if } i \in \mathcal{I}_{pos} \cup \mathcal{I}_{neg}. \end{cases} \quad (13)$$

The final prediction/score is obtained by running Eq. 7 again with $\tilde{S}_0$ as the initialization.

3.5 Efficient Incremental Inference

The above framework assumes that a full unlabeled testing set is available at the beginning of inference, a.k.a. transductive inference. In practice, testing data often arrive in a stream. This prohibits building a static graph at the beginning. Moreover, when the testing data grows infinitely, building an ever expanding graph is computationally prohibitive. In this section, we introduce two techniques to tackle this real-world challenge.

3.5.1 Dynamic Graph Construction for Incremental Inference

When testing data arrive in a stream fashion, a straightforward solution is to rebuild the entire KNN graph whenever a new sample is observed, but this quickly becomes computationally prohibitive at scale. Following the dynamic graph expansion strategy in [25], instead we update the graph incrementally to support inductive inference while still using the test-time manifold.

Specifically, given an existing adjacency matrix W built on the current node set, when a new test sample z_{N_u+1} arrives, we compute its similarities to all existing test nodes $\{z_i\}_{i=1}^{N_u}$, where $S_i = z_{N_u+1}^\top z_i$. We then insert the new node by reusing the existing graph structure: for each node i , we compare S_i with the weakest edge weight in the current neighborhood of i . If S_i is larger than the minimum neighbor weight of node i , we replace that weakest connection with an edge to the new node; otherwise, the neighborhood

of node i remains unchanged. Formally, the update can be written as

$$\begin{aligned} \forall i \in \{1, \dots, N_u\}, \quad S_i &= z_{N_u+1}^\top z_i \\ W_{i, N_u+1} &= S_i \cdot \mathbb{1}(S_i > \min_j W_{ij}), \\ W_{ij} &= W_{ij} \cdot \mathbb{1}(S_i > \min_j W_{ij}) \cdot \mathbb{1}(j \neq \arg \min_j W_{ij}). \end{aligned} \quad (14)$$

where $\min_j W_{ij}$ denotes the weakest connection in node i 's current neighborhood. This update inserts the new sample into the existing graph without recomputing k -NN neighborhoods for all nodes, thereby reducing the overall graph maintenance cost and enabling efficient streaming inference.

3.5.2 Graph Maintenance via Sliding Window

To prevent the test graph from growing indefinitely as more samples arrive, we further introduce a graph maintenance strategy. The motivation is straightforward: although the incremental update in Eq. (14) avoids rebuilding the entire graph from scratch, the number of test nodes may still increase continuously over time, which can eventually lead to excessive memory consumption. To address this issue, we impose a maximum capacity B on the number of active unlabeled test nodes, e.g. we use $B = 10000$ in the experiment.

When the current number of test nodes satisfies $N_u < B$, the graph is updated using Eq. (14) without any additional operation. Once the graph has already reached its maximum capacity, i.e., $N_u = B$, and a new test sample z_{B+1} arrives, we first perform one provisional dynamic update following Eq. (14), resulting in an intermediate graph W^+ . We then identify the most redundant historical test node with respect to the new sample by selecting the node with the largest cosine similarity:

$$r = \arg \max_{i \in \{1, \dots, B\}} z_{B+1}^\top z_i. \quad (15)$$

Intuitively, if one historical node is extremely similar to the incoming sample, then retaining both of them provides limited additional manifold information while unnecessarily enlarging the memory footprint. We therefore remove node u_r together with all its incident edges, yielding a pruned graph

$$W^- = \text{Delete}(W^+, r), \quad (16)$$

where $\text{Delete}(W^+, r)$ denotes deleting the r -th row and r -th column of W^+ . In the same way, the corresponding pseudo-label entry of node u_r is removed from the memory bank as well.

To avoid reconstructing the entire KNN graph after deletion, we only perform a local repair on the neighborhoods affected by removing z_r . Let

$$\mathcal{A}_r = \{i \mid W_{ir}^+ > 0 \text{ or } W_{ri}^+ > 0\} \quad (17)$$

denote the set of nodes whose neighborhoods were directly connected to the deleted node. For each affected node $i \in \mathcal{A}_r$, we recompute its similarity to the newly arrived sample,

$$\tilde{S}_i = z_{B+1}^\top z_i, \quad (18)$$

and reuse the same weakest-edge replacement rule to locally

rewire the graph:

$$\begin{aligned} \forall i \in \mathcal{A}_r, \\ W_{i, B+1}^* &= \tilde{S}_i \cdot \mathbb{1}(\tilde{S}_i > \min_j W_{ij}^-), \\ W_{ij}^* &= W_{ij}^- \cdot \mathbb{1}(\tilde{S}_i > \min_j W_{ij}^-) \cdot \mathbb{1}(j \neq \arg \min_j W_{ij}^-), \end{aligned} \quad (19)$$

All unaffected nodes keep their previous neighborhoods unchanged. In this way, the active test graph is always maintained within a fixed memory budget, while still preserving the advantages of incremental graph expansion. This fixed-window mechanism avoids unbounded graph growth and makes streaming inference more stable and memory-efficient in long-horizon deployment scenarios.

3.6 Overall Algorithm

We present the overall algorithm for unified iterative label propagation for vision-language models in Algorithm 1. In Fig. 2, we show an overview of the proposed framework for zero-/few-shot 3D understanding via graph propagation.

Algorithm 1 Graph Score Propagation for 3D understanding

- 1: **Input:** training data $\mathcal{D}_l$, testing data $\mathcal{D}_u$ and each class prompt templates $\mathcal{T}_c$
 - 2: **Output:** Final scores $\tilde{S}_T$
 - 3: **for** $\mathcal{D}_l$ and $\mathcal{T}_c$, **do**
 - Encode: $z_l = f(x_l)$ for training data and $\mathcal{P}_c = \{g(t_j(y_c))\}$ for prompt templates.
 - Prompt clustering: positive prototypes $\tilde{\mathcal{P}}$ (Eq. 10)
 - 4: **end for**
 - 5: **for** $\mathcal{D}_u$, **do**
 - Encode: $z_i = f(x_i)$ for testing data
 - Build graph to compute W (Eq. 5, 6)
 - Initialize score $S_0^{(i)}$ (Eq. 8)
 - Score propagate to compute S_T (Eq. 7)
 - Self-training to generate positive $\mathcal{I}_{pos}$ and negative prompts $\mathcal{I}_{neg}$ (Eq. 11)
 - Updated initial scores $\tilde{S}_0$ (Eq. 7)
 - Score propagate to compute $\tilde{S}_T$ (Eq. 7)
 - 6: **end for**
 - 7: **return** $\tilde{S}_T$
-

4 EXPERIMENTS

4.1 Evaluation on Out-of-Distribution Detection

4.1.1 Experimental setup

Datasets. We evaluate our method on the 3DOS benchmark [33], which is constructed from two widely used 3D object datasets, **ShapeNetCore** [60] and **ScanObjectNN** [61], and further extend the evaluation to several additional real-world datasets, including **Sydney Urban Objects** [62], **nuScenes** [63], and **S3DIS** [64]. ShapeNetCore contains 54 categories of synthetic CAD models covering common daily objects. ScanObjectNN is derived from the ScanNet dataset and consists of 2,902 real-world object point clouds from 15

categories. To more thoroughly assess our approach on real-world data, we additionally consider three LiDAR/scene benchmarks. The Sydney Urban Objects dataset provides LiDAR-scanned point clouds from 26 categories with 631 unique instances. The S3DIS dataset comprises 6 large-scale indoor areas with 271 rooms annotated into 13 semantic categories; we extract individual object point clouds using ground-truth segmentation masks, resulting in 8,931 objects. The nuScenes dataset contains LiDAR sequences captured in autonomous driving scenarios; we extract object point clouds using ground-truth 3D bounding boxes, yielding 2,205 instances from 9 categories. A detailed visualization of all datasets is provided in the supplementary material.

Evaluation protocol. We follow the 3DOS protocol [33] to construct ID/OOD splits. ShapeNetCore is partitioned into three disjoint subsets, SN1–SN3, each containing 18 categories. For each OOD setting, one subset is used as ID classes and the remaining two as OOD classes, yielding three test cases. ScanObjectNN is similarly divided into three 5-class subsets, SR1–SR3, which define three corresponding ID/OOD splits. For the Sydney Urban Objects dataset, categories are grouped into movable objects (e.g., truck, pedestrian) and non-movable objects (e.g., building, traffic sign). Movable classes are treated as ID and the rest as OOD, reflecting the focus of autonomous-driving perception on dynamic agents. S3DIS and nuScenes are adapted to object-level OOD detection as follows. In S3DIS, we retain instances with more than 2,048 original points, uniformly downsample them to 2,048 points, and designate foreground objects as ID and background objects as OOD. In nuScenes, we keep instances with more than 200 original points, upsample them to 1,024 points, and define four-wheeled vehicles (car, bus, truck, construction vehicle, trailer) as ID, with all remaining classes considered OOD. More preprocessing details are provided in [1]. Furthermore, for both few-shot and full-shot experiments, we only incorporated ID samples from the training set into the test samples, while excluding any OOD exemplars present in the training data.

Implementation Details. We use the pre-trained ULIP2 [23] model to extract point cloud features and text features. For the 3D encoder, we select Point-BERT [65]. In constructing the KNN graph, we set a universal K-value (N_k) of 10, connecting each text feature to its 10 nearest point cloud features. In Eq. 7, we set α to 0.5. The number of pseudo negative/positive samples, i.e. m in Sect. 3.4, is set to 5. Additionally, we use K-means clustering (N_c) to group the prompts for each category into 3 clusters.

Competing methods. We compare GSP++ with several representative OOD detection methods, including the full-shot baselines reported in 3DOS [33]. In the zero-shot setting, we consider recent methods DDCS [66], ZLaP [24], MCM [17], and NegLabel [52]. Unless otherwise stated, all zero-shot methods share the same 3D encoder as in [23]. We additionally report a simple baseline that directly uses cosine similarity between point-cloud features and text prototypes as the OOD score, denoted ULIP or ULIP2 depending on the underlying VLM backbone.

Evaluation metrics. We assess OOD detection performance using two standard metrics: AUROC and FPR95. AUROC measures the area under the receiver operating characteris-

tic curve, where higher values indicate better separability between ID and OOD samples. FPR95 denotes the false positive rate when the true positive rate is fixed at 95%, with lower values corresponding to stronger OOD detection performance.

4.1.2 Results on OOD Detection

Analysis of ScanObjectNN Results. From the ScanObjectNN results in Tab. 1, we draw two main observations. (i) In the zero-shot setting, the proposed extension GSP++ achieves the best average AUROC of 79.4%, clearly outperforming the cosine-similarity baseline ULIP2 as well as other recent zero-shot methods such as DDCS, MCM, NegLabel, and ZLaP. At the same time, its FPR95 is markedly lower than that of ULIP2 (76.7%), indicating that the additional components introduced in GSP++ further improve the separability between ID and OOD samples. Compared with the previous method GSP [1], GSP++ provides a consistent gain in AUROC while maintaining a comparable FPR95. (ii) When labeled training data are available, the family of GSP-based methods achieves performance that is competitive with or superior to fully supervised OOD detectors. In particular, the proposed GSP++(full-shot) attains the highest average AUROC of 92.1% on ScanObjectNN, exceeding the strongest supervised baseline Cosine proto by more than 8 AUROC points, while also substantially reducing FPR95. Compared with GSP(full-shot), GSP++ further improves AUROC with only a slight increase in FPR95, demonstrating that the new components bring measurable benefits even in the fully supervised regime.

Analysis of ShapeNetCore results. The ShapeNetCore results in Tab. 1 lead to two observations. (i) In the zero-shot setting, the performance gap between the strongest VLM-based baseline ULIP and graph-based methods is relatively small: ULIP already achieves an average AUROC of 79.9%, and the previous graph-based method GSP [1] attains a comparable AUROC of 80.4% with a different FPR95 trade-off. This narrow margin is largely due to the substantial overlap between the pre-training data of ULIP/ULIP2 (ShapeNet) and the downstream ShapeNetCore categories, which makes zero-shot OOD detection intrinsically easier on this benchmark. Nevertheless, the proposed GSP++ further improves upon GSP on this dataset, indicating that explicitly modeling the feature manifold is still beneficial even when the downstream distribution is close to the VLM pre-training data. (ii) With full supervision, GSP(full-shot) reaches an average AUROC of 98.2% and an FPR95 of 8.0%, clearly outperforming all other training-based baselines such as SubArcFace and Cosine proto. The GSP++ variants inherit this strong few-shot and full-shot behavior while further refining the AUROC–FPR95 trade-off over the original GSP, highlighting the effectiveness of the additional components introduced in this work.

Analysis of additional real-world datasets. The zero-shot OOD detection results on Sydney Urban Objects, nuScenes and S3DIS in Tab. 2 further support our conclusions on 3D real-world scenarios. (i) On Sydney and nuScenes, the proposed GSP++ consistently achieves the best AUROC, improving over the strongest non-graph baselines by around 8–11 AUROC points, while also yielding the lowest FPR95 on both datasets. Compared with the previous graph-based

TABLE 1

Evaluation of OOD detection for ScanObjectNN and ShapeNetCore datasets. Each column title for SR/SN indicates the chosen known class set and the rest sets of SR/SN are unknown.

	Method	ScanObjectNN								ShapeNetCore							
		SR3		SR2		SR1		Average		SN1		SN2		SN3		Average	
		AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓
Requiring training dataset	MSP [13]	88.1	67.3	80.6	84.0	73.7	80.3	80.8	77.2	74.3	82.8	80.0	78.1	89.7	52.2	81.3	71.0
	MLS [67]	89.4	53.8	83.4	73.1	76.4	75.3	83.0	67.4	72.0	80.8	83.9	64.1	89.8	40.5	81.9	61.8
	ODIN [68]	90.2	47.9	83.3	71.7	76.3	76.8	83.3	65.5	74.2	79.4	79.4	71.7	87.8	41.8	80.5	64.3
	Energy [69]	89.5	50.6	81.6	75.8	76.6	75.5	82.6	67.3	72.1	81.2	83.0	64.7	89.8	39.4	82.0	61.8
	GradNorm [70]	88.5	50.7	77.4	75.3	75.2	76.8	80.4	67.6	72.1	81.8	57.7	88.9	57.8	79.0	62.6	83.3
	ReAct [71]	90.3	48.9	82.4	75.8	75.4	77.6	82.7	67.4	73.7	79.4	89.6	52.1	95.0	27.2	86.1	52.9
	NF	88.0	47.7	80.6	68.2	75.6	81.4	81.4	65.8	81.5	72.5	71.1	78.0	91.0	49.6	81.2	66.7
	OE+mixup [72]	72.6	83.5	72.0	88.5	62.5	87.8	69.0	86.6	72.7	78.9	80.3	68.8	87.3	62.2	80.1	69.9
	ARPL+CS [12]	-	-	-	-	-	-	-	-	74.8	80.3	80.7	72.4	85.4	50.8	80.3	67.8
	Cosine proto	91.0	41.0	82.1	78.2	77.6	75.6	83.6	64.9	80.3	68.3	88.7	60.8	91.9	38.0	86.9	55.7
	CE	85.1	64.4	78.9	83.9	73.2	79.1	79.1	75.8	83.4	66.8	89.5	37.7	92.9	28.1	88.6	44.2
	SupCon [73]	-	-	-	-	-	-	-	-	80.9	75.5	83.5	68.2	85.1	45.1	83.2	62.9
	SubArcface [74]	87.1	61.3	78.9	76.9	73.7	81.4	79.9	73.2	79.0	81.2	82.9	60.3	89.1	32.8	83.7	58.1
	GSP(full-shot) [1]	90.1	33.0	93.0	30.5	92.2	34.4	91.8	31.6	97.9	10.1	97.9	7.9	98.8	6.0	98.2	8.0
GSP++(full-shot)	90.7	34.1	92.8	30.1	92.9	35.9	92.1	33.3	97.2	12.2	97.8	8.4	99.8	0.7	98.3	7.1	
Zero-shot	ULIP [10]	79.9	82.7	62.3	87.5	57.4	92.3	66.5	87.5	88.0	57.1	76.4	48.7	75.2	51.3	79.9	52.4
	ULIP2 [23]	65.7	80.8	76.4	71.2	77.4	78.0	73.2	76.7	85.7	52.5	84.1	44.3	67.6	74.4	79.1	57.1
	DDCS [66]	63.3	90.5	62.4	80.0	64.4	96.2	63.4	88.9	61.1	86.7	66.8	86.5	77.2	76.2	68.4	83.1
	MCM [17]	52.0	92.3	62.3	91.3	81.3	80.4	65.2	88.0	85.1	51.6	83.2	46.1	66.4	75.8	78.2	57.8
	NegLabel [52]	65.7	85.5	66.8	91.3	67.4	88.2	66.6	88.3	60.6	87.8	80.6	78.6	88.0	48.3	76.4	71.6
	ZLaP [24]	70.1	81.7	67.0	84.7	67.8	83.6	68.3	83.3	88.2	52.8	72.3	66.4	77.4	90.0	79.3	69.7
	GSP [1]	75.8	63.0	76.5	77.6	83.6	61.6	78.6	67.5	90.6	38.9	70.7	64.0	79.7	93.7	80.4	65.5
	GSP++	76.4	62.0	76.3	78.9	85.5	61.1	79.4	67.3	88.9	42.5	76.4	53.4	79.0	99.7	81.4	65.2

TABLE 2

Zero-shot OOD detection results on the Sydney Urban Objects dataset, nuScenes dataset and S3DIS dataset.

Method	Sydney		nuScenes		S3DIS	
	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓
ULIP [10]	63.4	81.7	53.2	93.3	53.1	95.1
ULIP2 [23]	66.2	83.7	64.2	93.0	62.5	92.3
DDCS [66]	61.8	87.3	62.2	91.2	62.5	92.5
MCM [17]	65.7	82.4	57.1	82.3	65.2	88.1
NegLabel [52]	58.4	92.7	69.7	81.6	55.4	85.4
ZLaP [24]	58.5	80.6	70.0	87.9	71.3	88.4
GSP [1]	72.5	68.9	80.8	80.7	77.6	94.0
GSP++	73.7	64.0	81.3	78.9	78.2	91.3

method GSP, GSP++ brings additional gains in AUROC together with noticeable reductions in FPR95, indicating that the new components introduced in this work further strengthen manifold-aware OOD detection under large distribution shifts between downstream data and VLM pre-training corpora. (ii) For S3DIS, GSP and GSP++ remain the top performers in terms of AUROC: GSP++ attains 78.2% AUROC, outperforming the best competing baseline ZLaP by nearly 7 points and exceeding MCM by about 13 points. At the same time, GSP++ slightly improves the FPR95 of GSP, although methods such as NegLabel achieve a lower FPR95 at the cost of substantially worse AUROC. This pattern suggests that our manifold-based approach is particularly effective at maintaining a strong global separation between ID and OOD samples across decision thresholds, while there is still room to further reduce false positives under very stringent operating points on complex indoor scenes.

Analysis for few-shot results. We further compare the few-shot OOD detection performance of GSP++ with the original GSP on ShapeNetCore under different shot settings, as illustrated in Fig. 3. Several consistent observations can be drawn across the three splits. First, GSP++ consistently outperforms GSP in the low-shot regime, particularly under extremely limited supervision. On SN1, GSP++ achieves noticeable AUROC gains over GSP when only one or two shots

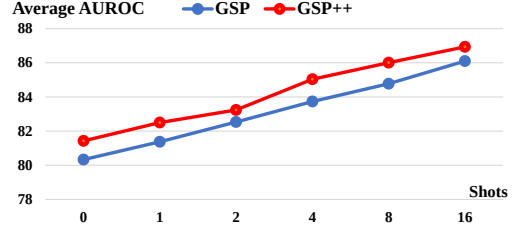


Fig. 3. Few-shot OOD detection on the ShapeNetCore dataset.

are available, indicating that the proposed enhancements enable more effective exploitation of scarce in-distribution samples. Similar trends are observed on SN2 and SN3, where GSP++ yields uniformly higher AUROC values across almost all shot numbers. Second, the performance gap between GSP++ and GSP becomes more pronounced as the number of shots increases from one to eight. This suggests that GSP++ benefits more from additional labeled samples, leading to a more reliable construction of the score propagation graph and more stable manifold estimation. Finally, when sufficient shots are provided, GSP++ maintains a consistently higher performance ceiling than GSP across all splits, demonstrating improved robustness and scalability in the few-shot setting. These results collectively validate that the proposed extensions in GSP++ significantly strengthen the original GSP framework, yielding superior and more stable OOD detection performance under varying levels of supervision.

4.2 Evaluation on 3D classification

4.2.1 Experimental setup

Datasets. We conduct experiments on the **ModelNet40** [75] and **ScanObjectNN** [61] datasets to evaluate the proposed GSP++ framework. ModelNet40 is a synthetic CAD model dataset containing 9,843 training samples and 2,468 testing samples across 40 object categories, which is widely used for benchmarking 3D shape classification and OOD detection.

ScanObjectNN is a real-world 3D point cloud dataset consisting of 2,902 samples from 15 object categories, featuring complex backgrounds and realistic noise, making it more challenging than synthetic datasets. We follow the same dataset construction, preprocessing, and evaluation protocols as ULIP2 [23] to ensure fair and consistent comparisons. **Implementation Details.** We use the pre-trained ULIP2 [23] model to extract point cloud features and text features. For the 3D encoder, we select Point-BERT [65]. In constructing the KNN graph, we set a universal K-value(N_k) of 10. In Eq. 7, we set α to 0.5. The number of pseudo negative/positive samples, i.e. m in Sect. 3.4, is set to 5. Additionally, we use K-means clustering(N_c) to group the prompts for each category into 3 clusters.

Competing methods. For zero-shot 3D classification, we benchmark our method against representative zero-shot 3D classification approaches that are commonly used in the ULIP2 [23] evaluation protocol, including PointCLIP [21], PointCLIPv2 [22], ReCon [76], CLIP2Point [37], and the OpenShape-based Point-BERT variant [65]. We further compare with ULIP [10] and ULIP-2 [23], which align 3D encoders with vision-language models and serve as strong recent baselines on both ModelNet40 and ScanObjectNN. In addition, we include ZLaP [24] and ECALP [25] as a recent competitive method for zero-shot 3D recognition.

For few-shot 3D classification, we compare GSP++ with both classical and recent 3D recognition baselines. Specifically, we include PointNet [77] and PointNet++ [78] as representative point-based architectures, CurveNet [79] as a strong geometry-aware backbone, and SimpleView [80] as a competitive multi-view baseline. We also compare with vision-language based few-shot transfer methods, including PointCLIP [21] and PointCLIPv2 [22], which leverage CLIP-style supervision for data-efficient 3D recognition. All methods are evaluated under the same few-shot protocol on ModelNet40 and ScanObjectNN, and we report Top-1 accuracy under 1-, 2-, 4-, and 8-shot settings.

Evaluation metrics. We evaluate 3D classification performance using Top-1 and Top-5 accuracy. Top-1 accuracy measures the fraction of test samples whose predicted class matches the ground-truth label, while Top-5 accuracy counts a prediction as correct if the ground-truth class appears in the top five ranked classes. We report both metrics for zero-shot and few-shot 3D classification on ModelNet40 and ScanObjectNN.

4.2.2 Results on 3D classification

Analysis for zero-shot Results. Table 3 reports the zero-shot 3D classification results on ModelNet40 and ScanObjectNN under the ULIP-2 evaluation protocol. We compare GSP++ with representative zero-shot baselines, including PointCLIP, PointCLIPv2, ReCon, CLIP2Point, and Point-BERT based approaches, as well as recent VLM-aligned 3D models ULIP and ULIP2. We further include ZLaP as an additional strong competitor. When pre-trained on ShapeNet, GSP++ achieves the best performance on both datasets. On ModelNet40, GSP++ reaches 75.4% Top-1 and 90.1% Top-5 accuracy, exceeding ULIP2 by 2.1% in Top-1 and remaining competitive in Top-5. On ScanObjectNN, GSP++ obtains 61.8% Top-1 and 89.9% Top-5 accuracy, improving over ULIP2 by 10.7% Top-1 and 3.9% Top-5, and

TABLE 3
Zero-shot 3D classification on ModelNet40 and ScanObjectNN.

Method	Pre-train Dataset	ModelNet40		ScanObjectNN	
		Top-1	Top-5	Top-1	Top-5
PointCLIP [21]	-	19.3	34.8	26.4	-
PointCLIPv2 [22]	-	63.6	85.0	36.4	-
ReCon [76]	ShapeNet	61.2	78.1	43.7	-
CLIP2Point [37]	ShapeNet	49.5	81.2	35.5	-
Point-BERT [65]	ShapeNet	70.3	91.3	-	-
Point-BERT [65]	Objaverse+ShapeNet	82.6	96.9	-	-
ULIP [10]	ShapeNet	60.4	84.0	43.2	83.0
ULIP2 [23]	ShapeNet	73.3	91.2	51.1	86.0
ZLaP [24]	ShapeNet	73.5	88.1	60.2	84.5
ECALP [25]	ShapeNet	74.8	90.2	56.6	89.5
GSP++	ShapeNet	75.4	90.1	61.8	89.9
Point-BERT [65]	Objaverse(noLVIS)+ShapeNet	83.9	97.6	-	-
Point-BERT [65]	Objaverse+ShapeNet+(2extra)	84.4	98.0	-	-
ULIP [10]	Objaverse(noLVIS)+ShapeNet	68.6	86.4	-	-
ULIP2 [23]	Objaverse(noLVIS)+ShapeNet	84.0	97.2	-	-
ZLaP [24]	Objaverse(noLVIS)+ShapeNet	85.8	97.8	-	-
ECALP [25]	Objaverse(noLVIS)+ShapeNet	86.0	98.0	-	-
GSP++	Objaverse(noLVIS)+ShapeNet	86.3	99.1	-	-

also outperforming ZLaP by 1.6% Top-1 and 5.4% Top-5. These gains indicate that the proposed GSP++ better leverages the test-time data manifold to refine class scores, which is particularly beneficial on the more challenging real-world ScanObjectNN benchmark. Under the stronger pre-training setting that jointly uses Objaverse (excluding LVIS samples) and ShapeNet, GSP++ further improves to 86.3% Top-1 and 99.1% Top-5 on ModelNet40. Compared with ULIP2 trained on the same pre-training data, this yields a 2.3% gain in Top-1 accuracy. Overall, the results demonstrate that GSP++ consistently strengthens zero-shot 3D recognition over strong VLM-based baselines across different pre-training datasets, with the most pronounced improvements observed on real-world data.

Analysis for few-shot Results. Table 5 reports the few-shot 3D classification results on ModelNet40 and ScanObjectNN under 1-, 2-, 4-, and 8-shot settings. GSP++ consistently achieves the best performance across all shot numbers and both datasets, with particularly large margins in the extremely low-shot regime. On ModelNet40, GSP++ attains 80.2% Top-1 accuracy in the 1-shot setting, which is substantially higher than the strongest baseline PointCLIPv2 at 64.2%. The advantage remains clear as more labeled samples are provided: GSP++ reaches 81.7%, 84.3%, and 84.6% under 2-, 4-, and 8-shot settings, respectively, consistently outperforming all competing methods. This indicates that GSP++ can effectively leverage a few labeled examples to rapidly improve recognition performance and quickly saturate to a strong accuracy level. On the more challenging real-world ScanObjectNN dataset, GSP++ shows even more pronounced improvements. It achieves 61.7% Top-1 accuracy with 1 shot per class, outperforming PointCLIPv2 by 24.3%. Similar gains are observed for 2-, 4-, and 8-shot settings, where GSP++ obtains 62.0%, 62.8%, and 63.4% Top-1 accuracy, respectively, consistently exceeding the best baseline by a large margin. Overall, these results demonstrate that GSP++ provides strong data efficiency and robustness for few-shot 3D classification, especially on real-world data with larger intra-class variations and background noise.

TABLE 4

The results of ablation study. For the OOD detection task, we report the results on ScanObjectNN and 3D classification on ModelNet40 and ScanObjectNN. The Score Prop.* represent Log-Scaled Score Propagation.

Method	Prompt. Clust.	Neg. Prompt	Score Norm	OOD detection		3D classification	
				AUROC↑	FPR95↓	ModelNet40	ScanObjectNN
Cosine Dist.	-	-	-	73.2	76.7	73.3	51.1
Score Prop.	-	-	-	76.4	72.6	74.0	56.5
Score Prop.	✓	-	-	77.2	73.6	74.2	57.4
Score Prop.	-	✓	-	77.3	71.4	74.8	60.3
Score Prop.(GSP)	✓	✓	-	78.6	67.5	-	-
Score Prop.*	✓	✓	-	79.0	67.8	75.1	61.5
Score Prop.*(GSP++)	✓	✓	✓	79.4	67.3	75.4	61.8

TABLE 5

Few-shot 3D classification results for top-1 accuracy on ModelNet40 and ScanObjectNN.

Method	ModelNet40				ScanObjectNN			
	1-shot	2-shot	4-shot	8-shot	1-shot	2-shot	4-shot	8-shot
PointNet [77]	28.4	32.3	55.0	65.2	18.7	25.4	25.7	34.8
PointNet++ [78]	40.1	55.8	72.4	77.6	27.0	32.3	40.9	47.8
CurveNet [79]	40.2	55.9	69.5	75.8	21.2	25.1	26.3	31.2
SimpleView [80]	28.9	35.4	58.0	69.3	23.2	25.0	29.1	32.5
PointCLIP [21]	53.1	68.0	76.7	81.1	30.2	42.4	46.0	50.1
PointCLIP V2 [22]	64.2	71.6	78.9	84.3	37.4	43.9	49.2	53.5
GSP++	80.2	81.7	84.3	84.6	61.7	62.0	62.8	63.4

4.3 Further Analysis

4.3.1 Results for different 3D encoders

Tab. 6 reports zero-shot OOD detection results on ScanObjectNN when varying the underlying 3D encoder. Across all encoders, the previous manifold-based method GSP and the proposed GSP++ consistently improve over the ULIP and ULIP2 baselines. With the relatively weak PointMLP encoder, ULIP attains an average AUROC of 66.6% and an FPR95 of 82.1%. GSP lowers the FPR95 to 73.3% while slightly increasing AUROC, and GSP++ further raises the average AUROC to 68.3% and reduces FPR95 to 72.1%. The gains become more pronounced for stronger encoders. With Point-BERT, ULIP reaches 66.5% AUROC and 87.5% FPR95, whereas GSP++ improves these numbers to 76.3% AUROC and 71.1% FPR95. A similar trend is observed for PointNet++, where GSP++ achieves 76.1% AUROC and 70.8% FPR95 compared with 66.0% AUROC and 84.9% FPR95 for ULIP. Even when ULIP2 is used as the pre-training pipeline, GSP and GSP++ still provide clear benefits: with

TABLE 6

Zero-shot 3D OOD detection for ScanObjectNN dataset with different 3D encoders.

Method	3D Encoder	SR3		SR2		SR1		Average	
		AUR↑	FPR↓	AUR↑	FPR↓	AUR↑	FPR↓	AUR↑	FPR↓
ULIP	PointMLP	65.2	79.0	55.0	94.5	79.8	72.8	66.6	82.1
GSP	PointMLP	56.7	86.4	68.2	74.3	78.7	59.1	67.8	73.3
GSP++	PointMLP	56.0	83.5	70.6	74.5	78.3	58.3	68.3	72.1
ULIP	Point-BERT	79.9	82.7	62.3	87.5	57.4	92.3	66.5	87.5
GSP	Point-BERT	64.9	82.3	78.4	65.1	80.9	68.9	74.7	72.1
GSP++	Point-BERT	67.5	80.0	78.8	64.6	82.3	68.7	76.3	71.1
ULIP	PointNet++	63.6	83.5	55.0	94.5	79.5	76.9	66.0	84.9
GSP	PointNet++	73.2	79.3	71.3	75.9	79.2	69.3	74.5	74.8
GSP++	PointNet++	72.4	78.7	74.4	68.7	81.6	65.0	76.1	70.8
ULIP2	Point-BERT	65.7	80.8	76.4	71.2	77.4	78.0	73.2	76.7
GSP	Point-BERT	75.8	63.0	76.5	77.6	83.6	61.6	78.6	67.4
GSP++	Point-BERT	76.3	65.0	75.2	80.0	85.5	61.1	79.0	68.7

TABLE 7

Zero-shot 3D classification on ModelNet40 and ScanObjectNN with different 3D encoders.

Method	Pre-train Dataset	3D Encoder	ModelNet40		ScanObjectNN	
			Top-1	Top-5	Top-1	Top-5
ULIP	ShapeNet	PointMLP	60.0	79.4	45.1	79.4
ZLaP	ShapeNet	PointMLP	60.6	76.2	47.4	75.7
ECALP	ShapeNet	PointMLP	61.4	79.2	46.5	78.5
GSP++	ShapeNet	PointMLP	61.7	80.0	48.5	88.1
ULIP	ShapeNet	Point-BERT	60.4	84.0	43.2	83.0
ZLaP	ShapeNet	Point-BERT	62.8	82.6	51.1	85.0
ECALP	ShapeNet	Point-BERT	61.7	85.1	52.8	86.1
GSP++	ShapeNet	Point-BERT	65.6	85.2	54.5	91.2
ULIP	ShapeNet	PointNet++	57.7	78.9	44.1	81.8
ZLaP	ShapeNet	PointNet++	60.1	73.5	44.3	73.7
ECALP	ShapeNet	PointNet++	60.0	79.5	47.8	84.3
GSP++	ShapeNet	PointNet++	59.3	76.7	48.9	86.3
ULIP2	ShapeNet	Point-BERT	73.3	91.2	51.1	86.0
ZLaP	ShapeNet	Point-BERT	73.5	88.1	60.2	84.5
ECALP	ShapeNet	Point-BERT	74.8	90.2	56.6	89.5
GSP++	ShapeNet	Point-BERT	75.4	90.1	61.8	89.9

Point-BERT, ULIP2 obtains 73.2% AUROC and 76.7% FPR95, while GSP++ increases AUROC to 79.0% and reduces FPR95 to 68.7%. These results demonstrate that manifold-aware OOD modeling is robust across diverse 3D encoders and that the additional components introduced in GSP++ yield consistent improvements over the original GSP.

As for 3D classification, the results are shown in Table 7 on ModelNet40 and ScanObjectNN when varying the 3D backbone. Across all evaluated encoders, GSP++ consistently delivers the best performance, demonstrating that the proposed score refinement is complementary to different 3D feature extractors. When using PointMLP as the 3D encoder, GSP++ improves ModelNet40 Top-1 accuracy to 61.7% and raises ScanObjectNN Top-1 accuracy to 48.5%, outperforming ULIP and ZLaP, and achieving a notably higher ScanObjectNN Top-5 accuracy of 88.1%. With the stronger Point-BERT encoder, the advantage becomes more pronounced: GSP++ reaches 65.6% Top-1 on ModelNet40 and 54.5% Top-1 on ScanObjectNN, substantially exceeding ULIP and also improving upon ZLaP on both datasets, while maintaining the best Top-5 performance on ScanObjectNN at 91.2%. Similar trends are observed with PointNet++, where GSP++ achieves the highest ScanObjectNN Top-1 accuracy of 48.9% among the compared methods. Finally, under the ULIP2 pre-training setting with Point-BERT, GSP++ attains the best overall results, achieving 75.4% Top-1 and 90.1% Top-5 on ModelNet40, as well as 61.8% Top-1 and 89.9% Top-5 on ScanObjectNN. Overall, these results indicate that GSP++ yields consistent gains across diverse 3D backbones and is particularly effective on the more challenging real-world ScanObjectNN benchmark.

4.3.2 Ablation study

In this section, we conduct ablative studies to scrutinize our GSP++ approach for 3D OOD detection and 3D classification tasks.

Effectiveness of the Designed Components. Table 4 reports a unified component-wise ablation of GSP++ on zero-shot OOD detection (ScanObjectNN) and zero-shot 3D classification (ModelNet40 and ScanObjectNN). Starting from the cosine-distance baseline, the OOD detection performance is 73.2% AUROC with 76.7% FPR95, and the zero-shot

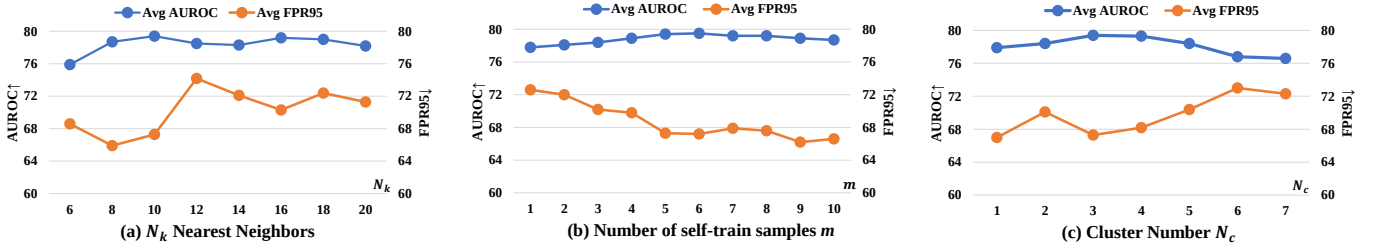


Fig. 4. OOD detection results from ScanObjectNN benchmark experiments: (a) Results of different number of N_k Nearest Neighbors in KNN. (b) Impact of varying self-train samples. (c) Evaluation of prompt clustering with different K-means cluster numbers.

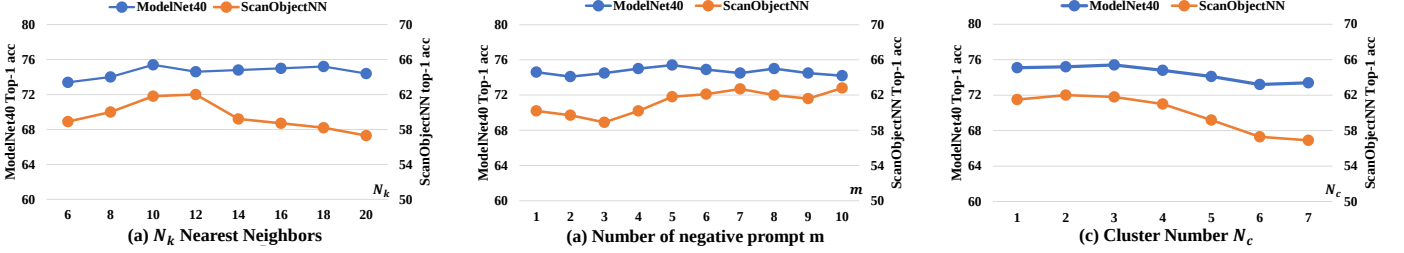


Fig. 5. 3D classification results from ModelNet40 and ScanObjectNN benchmark experiments: (a) Results of different number of N_k Nearest Neighbors in KNN. (b) Impact of varying self-train samples. (c) Evaluation of prompt clustering with different K-means cluster numbers.

classification accuracy is 73.3% on ModelNet40 and 51.1% on ScanObjectNN. Replacing direct similarity with score propagation consistently improves all tasks: AUROC rises to 76.4%, FPR95 drops to 72.6%, and classification accuracy increases to 74.0% on ModelNet40 and 56.5% on ScanObjectNN. This verifies that propagating scores on the test-time manifold provides more reliable predictions than using raw VLM similarity alone.

Based on score propagation, we then examine the effects of prompt clustering and negative prompts. Adding prompt clustering improves OOD AUROC from 76.4% to 77.2% and further raises classification accuracy to 74.2% on ModelNet40 and 57.4% on ScanObjectNN. Adding negative prompts gives a similar gain in AUROC, reaching 77.3%, but is more effective for reducing FPR95 (down to 71.4%) and brings a larger classification improvement, especially on ScanObjectNN, where the accuracy increases to 60.1%. Combining prompt clustering and negative prompts yields GSP, which further strengthens OOD detection and reaches 78.6% AUROC with 67.5% FPR95.

On top of GSP, we introduce log-scaled score propagation (denoted as Score Prop.*). This modification further improves OOD detection to 79.0% AUROC while maintaining a low FPR95 of 67.8%, and it also achieves 75.1% accuracy on ModelNet40 and 61.5% on ScanObjectNN. Finally, adding the proposed score normalization produces the full GSP++ model, which gives the best overall results: 79.4% AUROC and 67.3% FPR95 for OOD detection, together with 75.4% and 61.8% Top-1 accuracy on ModelNet40 and ScanObjectNN, respectively. The steady improvements across both tasks confirm that each component contributes positively, and that their combination is necessary to achieve the strongest overall performance.

Effectiveness of the Nearest Neighbors. Fig. 4(a) illustrates the impact of the number of nearest neighbors N_k used to construct the test-time graph in GSP++. As N_k increases from 6 to 10, the average AUROC steadily improves and the

average FPR95 decreases, indicating that an overly sparse graph limits the benefit of score propagation. When N_k lies between 10 and 18, both metrics fluctuate within a narrow range, showing that GSP++ is relatively robust to the choice of N_k in this region. The best trade-off is obtained at $N_k = 10$, where the model achieves the highest average AUROC together with one of the lowest FPR95 values. In contrast, very small or very large values of N_k lead to degraded performance, suggesting that both overly sparse and overly dense graphs can harm OOD separation.

Fig. 5(a) studies the effect of the number of nearest neighbors N_k used to build the test-time graph for zero-shot 3D classification. On both ModelNet40 and ScanObjectNN, increasing N_k from 6 to 10 leads to clear improvements in Top-1 accuracy, indicating that an overly sparse graph cannot fully exploit the manifold structure for score refinement. The performance on ScanObjectNN further peaks at $N_k = 12$, while ModelNet40 reaches its best result around $N_k = 10$. When N_k becomes larger than 12, ScanObjectNN accuracy drops noticeably and continues to decline as N_k increases, suggesting that an overly dense graph may introduce noisy inter-class connections in real-world data and weaken the discriminability of propagated scores. In contrast, ModelNet40 remains relatively stable for N_k between 12 and 18, reflecting its cleaner synthetic distribution, but also degrades when N_k is too large. Overall, these results demonstrate that GSP++ is moderately robust to N_k within a reasonable range, and we adopt $N_k = 10$ as the default setting to provide a strong and stable trade-off across both datasets.

Effectiveness of Self-Training Samples. We further study the influence of the number of self-trained samples used to construct negative prompts in GSP++. Using the ShapeNet-Core dataset, we vary the number of samples m that are assigned pseudo-negative labels for each ID class, as illustrated in Fig. 4(b). When m increases from 1 to 6, the average AUROC steadily improves, while the aver-

age FPR95 consistently decreases. This trend indicates that introducing a moderate amount of self-trained negatives is an effective way to exploit the data manifold and to refine the propagated scores. Beyond $m = 6$, both metrics begin to saturate and exhibit slight fluctuations, with a mild drop in AUROC for larger values of m . We attribute this behaviour to the accumulation of noisy pseudo-negative samples, which gradually offsets the benefit of additional self-training. Overall, $m = 6$ offers a good trade-off between performance and robustness.

Fig. 5 (b) investigates the influence of the number of self-trained samples m used to construct negative prompts for zero-shot 3D classification. Overall, ModelNet40 remains relatively stable across different choices of m , with only minor fluctuations in Top-1 accuracy, indicating that the synthetic distribution is less sensitive to the amount of pseudo-negative supervision. In contrast, ScanObjectNN exhibits a clearer dependency on m . When m increases from 1 to around 5–7, the Top-1 accuracy improves noticeably, suggesting that introducing a moderate number of self-trained negatives helps suppress confusing categories and refines the propagated class scores on the test-time manifold. When m becomes too small, the constructed negative prompts are insufficient to provide reliable counter-evidence; when m becomes large, the benefit saturates and the curve shows larger fluctuations, which we attribute to the accumulation of noisy pseudo-negative samples and potential confirmation bias. Based on this observation, we adopt a moderate value of m in our experiments to balance effectiveness and robustness.

Impact of Prompt Clustering We further investigate the effect of prompt clustering in GSP++ by varying the number of clusters N_c used in the K-means algorithm. Each cluster corresponds to a distinct prompt, so larger N_c yields a richer prompt set. As shown in Fig. 4 (c), using a single prompt per class gives an average AUROC below 78 and an FPR95 around 77. Increasing N_c from 1 to 3 steadily improves AUROC and slightly decreases FPR95; the best trade-off is obtained at $N_c = 3$, where GSP++ achieves its highest average AUROC and one of the lowest FPR95 values. When N_c is further increased beyond 4, both AUROC and FPR95 start to deteriorate, suggesting that overly aggressive clustering leads to redundant or noisy prompts and weakens the alignment between text and 3D features. These results indicate that using a moderate number of clusters is crucial for effectively exploiting prompt diversity, and we therefore adopt $N_c = 3$ as the default setting in GSP++.

Fig. 5(c) studies the effect of prompt clustering in GSP++ for zero-shot 3D classification by varying the number of clusters N_c in K-means. Each cluster corresponds to a distinct text prototype, and thus a larger N_c yields a richer prompt set. As shown in the figure, increasing N_c from 1 to 2–3 improves Top-1 accuracy on both ModelNet40 and ScanObjectNN, indicating that a moderate level of prompt diversification better captures intra-class linguistic variations and strengthens text–shape alignment. The peak performance is achieved around $N_c = 2$ on ScanObjectNN and $N_c = 3$ on ModelNet40. When N_c is further increased beyond 4, the performance on both datasets drops noticeably, with a sharper decline on ScanObjectNN. This suggests that overly aggressive clustering may fragment the prompt

TABLE 8
Computation cost of individual components on ShapeNetCore. We report the costs under three evaluation scales: 10% of the test set, the full test set, and the combined train-and-test set.

	Components	Sample size	FLOPs	Memory	Param.	Time Cost
Baseline cost	Dataload	515	0	48MB	0	1492ms
		5158	0	484MB	0	1531ms
		10000	0	922MB	0	1762ms
	Text Features Extract	515	191.0G	1.6G	13.7M	2459ms
		5158	191.0G	1.6G	13.7M	2459ms
		10000	191.0G	1.6G	13.7M	2459ms
	PC Features Extract	515	1307.6G	10.9G	171.9M	2553ms
		5158	1307.6G	11.3G	171.9M	25031ms
		10000	1307.6G	11.5G	171.9M	45560ms
Additional cost by GSP++	Prompt Clustering	515	1.1G	6.7MB	0	1657ms
		5158	1.1G	6.7MB	0	1657ms
		10000	1.1G	6.7MB	0	1657ms
	Build Graph	515	2.0G	11.0MB	0	72ms
		5158	20.1G	102.7MB	0	388ms
		10000	40G	198.1MB	0	537ms
	Score Propagation	515	0.03G	1KB	0	2ms
		5158	0.3G	14KB	0	5ms
		10000	0.3G	29KB	0	8ms

space and introduce noisy or less representative prototypes, which in turn weakens the discriminability of the refined class scores. Overall, the results highlight the importance of selecting a moderate cluster number, and we adopt $N_c = 3$ as the default setting to provide a strong and stable trade-off across both datasets.

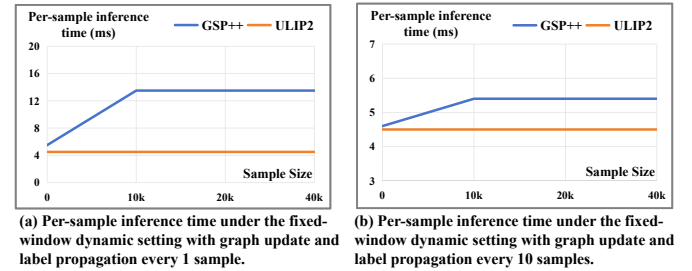


Fig. 6. Per-sample inference time comparison under the sliding-window dynamic setting. GSP++ performs graph update and label propagation after every incoming sample in (a) and after every 10 incoming samples in (b), and is compared with the ULIP-2 cosine-similarity baseline

4.3.3 Computation Cost of GSP++

Tab. 8 reports the computation cost of different components on ShapeNetCore. Overall, the cost is dominated by the basic VLM feature extraction, especially the point cloud encoder. For example, text and point-cloud feature extraction require 191.0 GFLOPs and 1307.6 GFLOPs, respectively, while the main GSP++ modules introduce much smaller costs. Even when the active test graph reaches the sliding-window capacity $B = 10000$, prompt clustering, graph construction, and score propagation only require 1.1, 40.0, and 0.3 GFLOPs, respectively. Their total cost is about 41.4 GFLOPs, which is only a small fraction of the feature extraction cost. The memory overhead is also limited, requiring about 205 MB in total compared with more than 13 GB used by feature extraction.

In terms of runtime, GSP++ also brings limited overhead. At the sliding-window capacity $B = 10000$, prompt clustering, graph construction, and score propagation take 1657 ms, 537 ms, and 8 ms, respectively, which are much smaller than the 45560 ms consumed by point-cloud feature extraction. Moreover, for streaming inference, the dynamic graph update only costs 10.4 MFLOPs, 42 KB memory, and 1 ms when maintaining a graph with $B = 10000$ active nodes. This shows that the sliding-window dynamic strategy can support continuous sample arrival with negligible per-sample update cost, without repeatedly rebuilding the full graph. These results demonstrate that GSP++ achieves test-time adaptation with very low additional computational cost.

Fig. 6 further provides an intuitive per-sample runtime comparison between GSP++ and the ULIP-2 cosine-similarity baseline under the sliding-window dynamic setting. The per-sample time of ULIP-2 mainly comes from the point-cloud encoder, which is about $45.56\text{s}/10000 \approx 4.5\text{ms}$ per sample. For GSP++, the additional cost consists of dynamic graph update and score propagation. When label propagation is performed after every incoming sample, the per-sample time is approximately $4.5\text{ms} + 8.0\text{ms} + 1.0\text{ms} = 13.5\text{ms}$ at the sliding-window capacity, where 8.0ms and 1.0ms correspond to score propagation and dynamic graph update, respectively. When the graph is updated every 10 incoming samples, these two costs are amortized over the 10 samples, leading to about $4.5\text{ms} + 8.0\text{ms}/10 + 1.0\text{ms}/10 = 5.4\text{ms}$ per sample. Therefore, as shown in Fig. 6 (b), the amortized per-sample runtime of GSP++ becomes very close to the ULIP-2 baseline, while still enabling test-time graph-based score propagation.

4.3.4 Pseudo-Label Quality

To further analyze the impact of pseudo label selection, we vary m from 1 to 50 and examine the correct rate on selected pseudo labels on 3D classification task. As shown in Fig. 7, the correctness of selected pseudo labels is generally high when m is small, and gradually degrades as m increases. On ModelNet40, the correct rate remains stable at around 90% for the 0-shot setting and even exceeds 94% for the 1-shot and 4-shot settings when $m \leq 10$. This indicates that the proposed confidence-based selection indeed concentrates on reliable samples, and a small pseudo-label set can provide clean supervision signals for subsequent propagation. However, when m grows beyond 20, the correct rate drops noticeably, and the decline becomes more significant at $m = 30$ and $m = 50$. This trend suggests that expanding the pseudo-label pool inevitably brings in harder and more ambiguous samples whose predicted classes are less reliable, which in turn increases the noise injected into the propagation process.

A similar phenomenon is observed on the more challenging ScanObjectNN benchmark. The overall correct rates are lower than those on ModelNet40, reflecting the larger domain gap and real-world noise in scanned point clouds. Nevertheless, few-shot supervision consistently improves pseudo-label quality: 1-shot and 4-shot settings yield higher correct rates than 0-shot for most m values, demonstrating that even a handful of labeled samples can calibrate the

score distribution and reduce systematic errors in pseudo-label selection. Meanwhile, the performance peaks at moderate m values and then decreases steadily as m increases, indicating that overly aggressive selection is detrimental on ScanObjectNN. In practice, these results justify using a relatively small m to trade off between coverage and label noise, and they also motivate our design of soft pseudo-label assignment, which mitigates the adverse impact of imperfect pseudo labels.

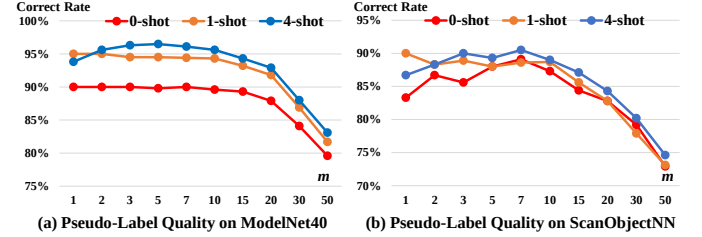


Fig. 7. Correct rates of pseudo samples with different selection size m

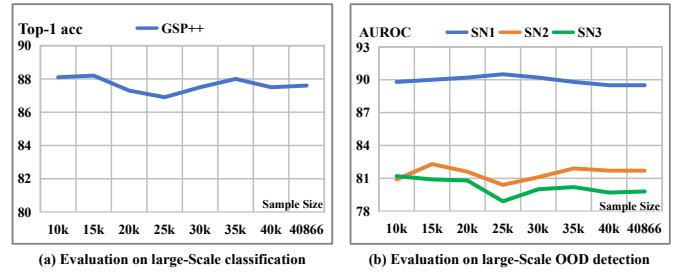


Fig. 8. Evaluation on large-scale dataset for two downstream tasks.

4.3.5 Efficient Incremental Inference

Dynamic Graph Expansion for OOD Detection. Fig. 9 illustrates the dynamic graph construction behavior for OOD detection on ScanObjectNN, while also comparing it with the cosine-similarity baseline. Several observations can be drawn. First, the baseline, which directly computes AUROC from cosine similarity, remains relatively stable across different sample ratios, whereas the performance of GSP++ in the dynamic setting is more sensitive to the amount of available test data, since its score propagation quality depends on the completeness of the evolving test graph. This phenomenon is particularly evident on SR1 and SR3, where the dynamic GSP++ curve exhibits larger fluctuations at the early stage due to the incomplete manifold structure and less reliable neighborhood relations. Second, as more test samples are progressively inserted, the performance of GSP++ improves consistently and gradually becomes highly competitive with, or even superior to, the full-sample counterpart. On SR3, the dynamic GSP++ result rises rapidly after the initial stage and remains clearly above both the dynamic and full-sample baseline, even slightly surpassing the full-sample GSP++ result in the middle and later stages. On SR2, the dynamic GSP++ curve approaches the full-sample result after only a small number of updates and then stays very close to it, indicating that the proposed update mechanism can recover a stable graph structure efficiently in relatively easier scenarios. On the more challenging SR1 split, although the dynamic curve starts much lower and

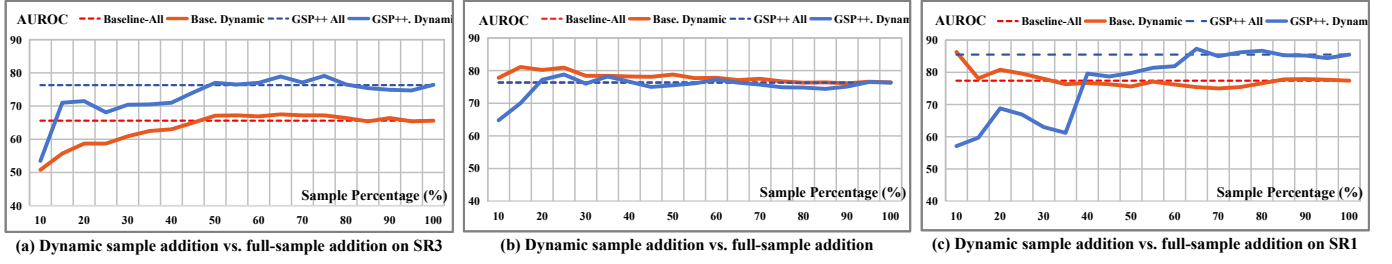


Fig. 9. Dynamic graph construction experiments for OOD detection on ScanObjectNN. The baseline computes AUROC directly using cosine similarity between visual and textual features. “Dynamic” reports the performance over all observed samples as new test samples are incrementally added, whereas “All” reports the performance when all test samples are added at once.

fluctuates noticeably at the beginning, it improves substantially as the graph expands, eventually approaching and even slightly exceeding the full-sample GSP++ result, while maintaining a clear advantage over the cosine-similarity baseline. Overall, these results demonstrate that the proposed dynamic graph update mechanism can effectively incorporate newly arrived test samples and progressively refine the graph structure, enabling strong OOD detection performance without rebuilding the full graph from scratch.

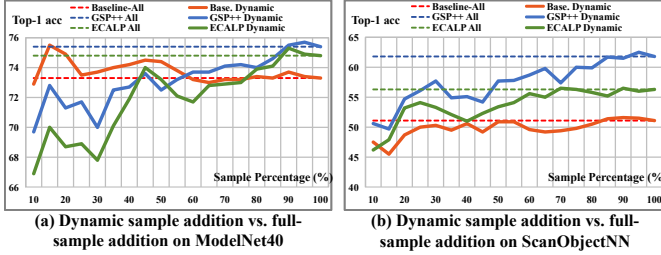


Fig. 10. The Dynamic Graph Expansion experiments for classification on ModelNet40 and ScanObjectNN. The baseline computes accuracy directly using cosine similarity between visual and textual features. “Dynamic” reports the performance over all observed samples as new test samples are incrementally added, whereas “All” reports the performance when all test samples are added at once.

Dynamic Graph Expansion for Classification.

Fig. 10 illustrates the dynamic graph expansion behavior for 3D classification on ModelNet40 and ScanObjectNN. To compare with an existing adaptive graph method, we further adapt ECALP to the same streaming protocol and sample arrival order. When only a small fraction of test samples is available, GSP++ may occasionally underperform the cosine-similarity baseline or exhibit a clear gap from its full-sample result. This is mainly because the test manifold is incomplete at the beginning of dynamic inference, resulting in insufficient or unreliable neighbors for score propagation. Moreover, we keep the graph construction and propagation hyperparameters fixed across all sample ratios rather than tuning them specifically for extremely small graphs. As more samples are inserted, GSP++ steadily improves and approaches its full-sample performance. On ModelNet40, GSP++ outperforms ECALP at most stages and reaches a higher final accuracy of 75.4%, compared with 74.8% for ECALP. The advantage is more pronounced on ScanObjectNN, where GSP++ achieves 61.8%, compared with 56.6% for ECALP. Both methods support incremental graph expansion and exploit the evolving test manifold; therefore, the improvement does not arise from a fundamental difference in whether historical samples can

be updated. Instead, GSP++ incorporates several components specifically designed for 3D point-text representations. Prompt clustering preserves diverse textual anchors, while self-training introduces informative pseudo-positive and pseudo-negative samples into the graph. Moreover, soft score calibration, score normalization, and log-scaled propagation regulate the influence of pseudo samples and stabilize propagation as the graph expands. These designs are particularly beneficial on ScanObjectNN, whose real-world point clouds contain stronger noise and distribution variations. Consequently, GSP++ maintains more reliable classification performance throughout dynamic graph expansion while retaining the efficiency of incremental inference.

Sliding-window Dynamic Update for Large-scale Dataset.

To further evaluate the scalability of GSP++ under long-horizon streaming inference, we conduct a large-scale experiment on the ShapeNetCore split provided by 3DOS [33], where the training and test samples are combined, resulting in 40,866 point clouds in total. As shown in Fig. 8, the proposed sliding-window dynamic update maintains stable performance when the number of processed samples increases from 10k to the full 40,866 samples. For 3D classification, the Top-1 accuracy only exhibits minor fluctuations and does not show a clear degradation trend as the stream becomes longer. For OOD detection, the AUROC curves on SN1, SN2, and SN3 also remain relatively stable, with only moderate variations during the intermediate streaming stages. These results indicate that retaining a fixed-size active graph with $B = 10000$ is sufficient to preserve useful test-time manifold information, while avoiding the continuously increasing memory cost caused by unbounded graph expansion. Therefore, the sliding-window strategy makes GSP++ more suitable for large-scale and streaming deployment scenarios.

5 CONCLUSION

In this work, we presented GSP++, a novel training-free approach for unified 3D point cloud understanding that effectively leverages Vision-Language Models through graph score propagation. By constructing task-specific graphs that incorporate class prototypes, testing samples, and optional few-shot labeled samples, our method exploits the underlying manifold structure of test data to improve both 3D classification and OOD detection. Compared with the original GSP, GSP++ further enhances score propagation with prompt clustering, self-training-based pseudo-label

samples, and log-scaled score propagation with score normalization, leading to more robust and accurate predictions. We also developed mechanisms to enable efficient incremental inference. Extensive experiments on multiple 3D benchmarks demonstrate the effectiveness, flexibility, and scalability of GSP++ under zero-shot and few-shot settings, as well as its practicality under dynamic graph expansion. Future work could explore more efficient large-scale graph construction and maintenance and stronger integration with emerging 3D foundation models for broader open-world 3D understanding tasks.

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Tiankai Chen is a Ph.D. student with the School of Computing and Artificial Intelligence, Southwest Jiaotong University, China, and he is under the supervision of Prof. Tianrui Li and Dr. Xun Xu. His research interests include 3D point cloud, large language model. He has published papers on ICCV'25, ICLR'26. Now he is a visiting student in Institute for infocomm research(I²R), A*STAR, Singapore.



Zhen Liang is a Ph.D. student at the University of Chinese Academy of Sciences (UCAS), China, under the supervision of Prof. Enhai Liu and Prof. Rujin Zhao. His research interests include stereo vision, foundation model-based adaptation, and spatial vision measurement. He is currently a visiting student at the Institute for Infocomm Research (I2R), A*STAR, Singapore.



Yushu Li is a Ph.D. student in Electrical and Computer Engineering at the University of British Columbia (UBC), Canada, advised by Prof. Xiaoxiao Li. His research interests include large language models, reinforcement learning, inference-aware scaling, and test-time adaptation.



Adam Goodge received a MSci. degree from the University of Birmingham in 2017 and PhD degree from the National University of Singapore in 2022. He is now with the Institute for Infocomm Research, I2R, A*STAR. His research interests include vision-language modeling, test-time adaptation, and robust learning, developing methods to enhance model reliability, generalization, and performance under distribution shift in real-world environments.



Tianrui Li is a Professor and Ph.D. Supervisor, and serves as the Dean of the School of Computing and Artificial Intelligence, Southwest Jiaotong University, China. He received the B.S., M.S., and Ph.D. degrees from Southwest Jiaotong University in 1992, 1995, and 2002, respectively. His research interests include artificial intelligence, data mining and knowledge discovery, cloud computing and big data, and granular computing/rough sets.



Fei Teng is a Professor and Ph.D. Supervisor, and serves as Associate Dean of the School of Computing and Artificial Intelligence, Southwest Jiaotong University, China. She received the Ph.D. degree from École Centrale de Paris, France, and was a visiting scholar at the University of Leeds, U.K. Her research interests include big data intelligence, cloud/service computing, and industrial big data.



Xun Xu is a Senior Scientist at the Institute for Infocomm Research (I²R), A*STAR, Singapore, where he leads the AI for Semiconductor research group. He received the Ph.D. degree from Queen Mary University of London in 2016 and was a Research Fellow at the National University of Singapore from 2016 to 2019. His research interests include data-efficient learning, robust AI, and generative/multimodal LLMs with applications to 3D data and industrial visual inspection.



Xulei Yang is a Principal Scientist and Group Leader at the Institute for Infocomm Research (I²R), A*STAR, Singapore, and was previously the Head of Research at YITU Technology Singapore. He received the Ph.D. degree from Nanyang Technological University (NTU) in 2007. With over 16 years of R&D experience in deep/machine learning for computer vision and healthcare, he has published more than 100 scientific papers and international patents in deep learning, 3D vision, and medical imaging. He is

an IEEE Senior Member and a Kaggle Competition Master.